

Backpack containing
life-support systems

Pressure
helmet with
protective visor

Gloves lock
onto suit
at wrists

Main suit in
two parts
that join
at waist

White material
reflects sunlight
and heat

Layered fabrics
trap air and
protect wearer

Living in space

Keeping astronauts fit and healthy as spaceflights have become longer and more complex is a huge challenge, and has required considerable inventiveness. And not only do astronauts have to live and work in cramped conditions, they also have to be kept safe when they leave their spacecraft.

Controls for
life-support systems

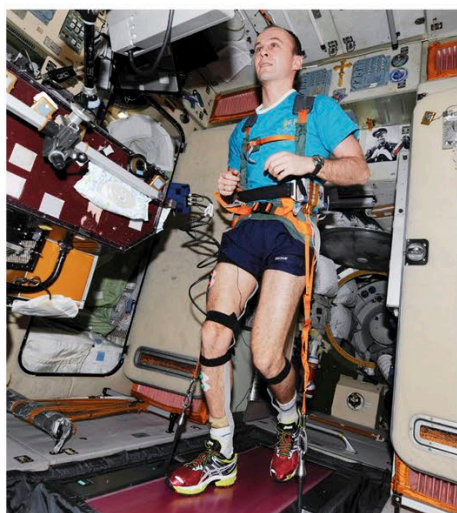
SPACE CLOTHES

The first astronauts wore protective spacesuits throughout their missions but, by the mid-1960s, safety improvements and larger spacecraft allowed them to spend most of their time in more comfortable flight suits. By the 1990s, the spacesuits worn by NASA astronauts for working outside the Space Shuttle (left) had developed into self-contained personal spacecraft.

Astronaut Cady Coleman
washes her hair with a
water-saving shampoo.

WOW!

All waste water on the ISS is recycled by passing through a series of filters and chemical reactions. It ends up cleaner than our drinking water on Earth.



KEEPING FIT

As missions have lengthened from mere days to months, it's become more important to keep astronauts in shape for their return to Earth. Weightless conditions in zero-gravity space weaken muscles and bones, so astronauts take a variety of supplements and exercise regularly, often using elastic straps to mimic the pull of gravity.

KEEPING CLEAN

Personal hygiene in space can be a problem—water is too precious to waste and, if it is sprayed around, it forms weightless drops that interfere with delicate electronic systems on board. Instead of showers, astronauts use small pouches of liquid soap, water, and “no rinse” shampoos.